

BRIAN ANTHONY EPSTEIN

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Education

Georgia Institute of Technology

Atlanta, Georgia

Ph.D. Mechanical Engineering, GPA: 4.00 — Research Area: Autonomy and Controls

August 2022 – Fall 2026

M.S. Mechanical Engineering, GPA: 4.00 — Controls & Robotics, Mechatronics & Design

August 2022 – May 2024

B.S. Mechanical Engineering, GPA: 3.96 — Minor: Robotics, Honors Program, Highest Honors

August 2018 – May 2022

Research Experience

Georgia Tech Aerial Robotics Experimental Autonomy Lab

August 2022 – Present

Graduate Researcher - (PI: Dr. Jonathan Rogers)

Atlanta, Georgia

Solar-Powered Autonomous Seaplane (Dissertation)

August 2023 – Present

- Developing a 5.0kg solar-powered seaplane UAV for ocean monitoring at ~\$1k versus ~\$50k/day chartered vessels.
- Invented a spanwise rotating wing giving ~22% solar gain and capsize self-righting with a single actuator.
- Fabricated and flight-tested the airframe in foaming filament, carbon spar, and fiberglass layup on a 3-day build cycle.
- Waterproofed the airframe for open-water landings with conformal coating, potted penetrations, and gasketed hatches.
- Sized the 3D printed hull to metacentric height targets, validated statically stable under full load in open water.
- Integrated thin-film solar, MPPT, and BMS with a custom PX4 tilt-wing tracking module.
- Built the VSPAERO model setting PX4 tuning and the 150W/18 m/s cruise budget, validated across flight tests.
- Authored a JSBSim airframe model, verifying every flight-software change in PX4 SITL before it reached the vehicle.
- Developed an MDP planner for fly-vs-float energy decisions under stochastic wind and solar, solved by dynamic programming, cutting failure rate 30% and battery capacity from 450 to 250 Wh against a tuned threshold heuristic.

University Consortium for Applied Hypersonics (Fault-Tolerant Estimation)

August 2025 – June 2026

- Developed state estimation for fault-tolerant control of hypersonic vehicles in Simulink and C++.
- Analyzed hypersonic flight test data to characterize vehicle behavior and inform estimator design.

Autonomous Drone Capture (Quadcopter Interceptor)

August 2022 – May 2023

- Designed and built a 500mm quadcopter with a custom basket to capture a target drone mid-air in a Vicon flight lab.
- Built a Python pursuit-intercept stack (A*, RRT*, B-spline, pure-pursuit) over a PX4-ROS2 offboard link.

Work Experience

Georgia Institute of Technology

Spring 2024 & Spring 2025

Flight Test Engineering Course Assistant

Atlanta, Georgia

- Served as primary point of contact for ~60 students per semester on lab, hardware, and flight-dynamics questions.
- Stood up shared simulator usage procedures in the course's inaugural semester, eliminating lab scheduling collisions.
- Developed Simulink flight-dynamics models integrated with X-Plane 12 for real-time aircraft-state visualization.

JLG Industries

May 2022 – August 2022

Engineering Research and Development Intern

Hagerstown, Maryland

- Implemented depth perception and plane detection with RANSAC, Open3D, and OpenCV on a RealSense D435.
- Prototyped autonomous powerwashing demonstration using 6DOF robotic manipulator, JLG hardware and ROS2.

Eaton Corporation

May 2021 – August 2021

Robotic Manufacturing Intern – Operations Leadership Development Program

Hastings, Nebraska

- Programmed Fanuc robot and CNC lathe to automate gear machining rework resulting in 75% cycle time reduction.
- Developed and implemented system for automotive gear machining bottleneck prioritization leading to \$40,000 savings.
- Championed continuous improvement for CNC operation and forging material handling leading to \$12,000 cost out.

East West Manufacturing

January 2020 – May 2020

Engineering Intern

Atlanta, Georgia

- Designed prototype eddy-current brake components, generated GD&T engineering drawings, and fabricated parts.
- Built two lab capabilities from scratch: a vacuum thermoforming table and a hemi-anechoic acoustic test chamber.
- Authored engineering drawings for a wide variety of parts used by both US and overseas machinists.

Lockheed Martin

May 2019 – August 2019

Manufacturing Engineering Intern

Ocala, Florida

- Cut missile assembly defect rate with a smart locking fixture built in-house for \$16,000 under the external quote.
- Integrated Arduino controlled smart fixture with existing torque driver system, minimizing defect escapes.

Selected Projects

Combat Robotics: Chonkiv | *RoboJackets (Georgia Tech Competitive Robotics Club)*

August 2018 – Present

- Designed the shuffling locomotion earning NHRL's weight bonus, enabling extra armor and weapon energy.
- Designed a full-body shell spinner to keep drivetrain and electronics functional after hits that deform the shell.
- Packaged motors, ESCs, battery, and harness in a sealed high-shock volume with retained connectors and strain relief.
- Modularized drive pods and weapon assembly for field repair so a subassembly easily swaps out between matches.
- Specified waterjet, CNC, and heat-treated parts with tolerances holding through hardening at outside vendors.
- Diagnosed and repaired structural, drivetrain, and electrical damage on-site between matches across a 17-3 record.
- 2x NHRL champion, first collegiate featherweight champion, ranked 1st in the world for a year and top 5 for two.

BattleBox — Combat Robotics Test Arena | *RoboJackets*

August 2023 – 2026

- Wrote and pitched the funding proposal that won a Georgia Tech Student Foundation grant to build the arena.
- Converted a shipping container into a containment arena for live 12lb and 30lb combat robot testing, now in team use.
- Specified internal armor and high-speed camera instrumentation to contain debris and estimate stored weapon energy.

Skills and Coursework

Hardware: Rapid Prototyping and Fabrication (Waterjet, Laser Cutter, 3D Printer, CNC/Manual Machine Tools, Soldering), Wiring harnesses, Environmental sealing, Payload Integration, Microcontroller Integration

Software: Python, MATLAB/Simulink, C/C++, ROS2, PX4, Gazebo, JSBSim, LabView, CAD (Solidworks, Autodesk Inventor, Fusion360, PTC Creo), OpenVSP/VSPAERO

Relevant Coursework: Advanced Control System Design and Implementation, Intro to Artificial Intelligence, Linear and Nonlinear Controls Systems, Dynamics of Mechanical Systems, Modeling and Simulation Design, Intelligent Cyberphysical Systems, Rotorcraft Aeromechanics, Machine Design

Publications

B. Epstein and J. D. Rogers, "Optimal flight planning for marine mammal surveillance from a seaplane UAV," *J. Aircraft*, in press.

B. Epstein and J. D. Rogers, "Conceptual design of a low-altitude solar powered seaplane for oceanic monitoring," in *Proc. AIAA AVIATION Forum and ASCEND*, 2024, AIAA 2024-3751.

D. Ward, **B. Epstein**, L. Tiziani, and F. L. Hammond III, "Optimal design of a mechatronic lever arm for pneumatic exoskeleton: Design and validation," in *Proc. Design of Medical Devices Conf.*, 2021.

Leadership

Woodruff Graduate Leaders Fellow | *Woodruff School of Mechanical Engineering*

Fall 2025

- Selected to the inaugural 22 fellow cohort from the College of Engineering for a graduate student leadership program.
- Represented graduate student concerns to administration, proposing solutions adopted into the program's continuation.

Undergraduate Research Mentor | *Aerial Robotics Experimental Autonomy Lab*

2023 – Present

- Mentored over 25 undergraduate researchers across machine design, manufacturing, controls, and flight testing.
- Mentees have since advanced to Sandia National Laboratories, Bell, and aerospace-robotics PhD programs.

RoboJackets Combat Robotics | *Georgia Tech Competitive Robotics Club*

August 2018 – Present

- Serve on the board of RoboJackets Inc., the 501(c)(3) managing sponsorship funds and winnings for the club.
- Held Mechanical Training Lead, 3lb Mentor, Sponsorship Chair, Team Lead, and Team Mentor roles across 8 years.
- Trained incoming members in machine design and fabrication, sustaining the program through membership turnover.

Emerging Leaders | *Member 2018–19, Board of Advisors 2019–21*

2018 – 2021

- Completed a year-long leadership program, executing member-led service projects with a 30-student cohort.
- Returned for two years on the Board of Advisors, developing each incoming class of 30 first-year students.
- Shaped cohort selection, curriculum, and the fall retreat, coaching members through projects they planned and ran.